PROFESSIONAL CV · 2026

Bing Gao

Computational Mechanics Engineering Student / Fluid Simulation, Thermal Systems & Engineering Software

gaobing1230@gmail.com   +33 6 30 65 97 65   Paris La Defense, France   LinkedIn   Code portfolio

Engineer with nearly 7 years in mobile software development and technical leadership, now applying experience from a product serving more than 2 million users to fluid simulation, finite-element analysis, thermal modelling, and numerical verification while completing a French engineering degree.

Nearly 7 years	2M+	2027
Software delivery and technical leadership	Users reached by a product built from scratch	Engineering degree expected

01

Experience

Engineering and product delivery

Aerodynamics Engineer

Sep 2025 — Present

Vinci Eco Drive, ESILV Formula Student · Paris La Defense, France

- Led thermal-system modelling for the battery, inverter, and motor, coupling hydraulic, radiator, and transient coolant models to screen the E3 cooling architecture before procurement.
- Developed cooling-duct concepts within the aerodynamic package and used an OpenFOAM fan/core surrogate to qualify the numerical method before detailed vehicle fluid simulation.

3D Modelling Intern

Apr — Aug 2026 · 17 weeks

Felisiak Engineering & Development · Paris, France

- Reconstructed the ESA Space Rider from public blueprints for a launch-campaign review application and delivered separate review and animation-ready assets for Unity and Blender.
- Built a scripted Blender audit, rollback, and export workflow and prepared the engineering report, defense poster, and browser-ready GLB.

Mobile Application Development Manager

Aug 2016 — Mar 2023

Inkeverse Group Ltd · Beijing, China

- Led Lao You from inception to more than 2 million users; built the product from scratch, wrote 70% of its initial core code, and helped establish it as a revenue source.
- Independently developed voiceprint-recognition algorithms and real-time gift-rendering effects for live broadcasts.
- Designed a Ruby-based delivery pipeline that automated testing, packaging, and distribution across Android and iOS.
- Implemented code and resource obfuscation plus certificate pinning to increase resistance to reverse engineering and traffic interception.
- Integrated React Native and built a hotfix mechanism for controlled runtime updates.

02

Selected work

Methods, evidence, and stated limits

Formula Student Cooling System & OpenFOAM Surrogate

Vinci Eco Drive · 2025–2026

- Coupled vendor fan, radiator, and pump curves with an 80-cell one-dimensional finite-volume coolant model; the hydraulic operating point reached 10.03 L/min.
- Identified that the passive E3 architecture violated the inverter temperature boundary before procurement and detailed vehicle simulation, documenting a rejection decision and replacement architecture paths.
- Qualified a fan-jump and porous-core OpenFOAM surrogate on four meshes from 7,440 to 3,481,920 cells; fine-to-extra-fine changes were 0.950% in flow and 0.0145% in pressure loss.
- Limit: numerical screening and surrogate-method qualification, not installed-vehicle airflow, heat transfer, hardware, or track validation.

F1 External-Aerodynamics RANS Pilot

Independent study · 2026

- Built a 4.35-million-cell OpenFOAM 14 half-car RANS baseline and completed a 23-variant pilot campaign across geometry, boundary conditions, and modelling sensitivities.
- Automated CAD variant staging, snappyHexMesh, steady RANS, validity gates, component-force extraction, and result archiving; retained nine valid sensitivity cases and preserved two diverged roughness runs.
- Kept material mesh sensitivity visible and excluded absolute aerodynamic performance, yaw, wind-tunnel, and physical-correlation claims.

FlowLab & FlowROM

Independent study · 2026

- Implemented a dependency-free D2Q9 BGK lattice-Boltzmann solver shared by Node.js and the browser; validated three grids at Re=100 against canonical lid-driven-cavity data.
- Generated 480 velocity fields, using 320 snapshots for training and 160 snapshots across four complete holdout cycles for POD/DMD evaluation.
- Achieved 0.123% rank-8 POD holdout error with 48.8× compression and 0.100% DMD full-state holdout error; the stated limit is controlled two-dimensional internal flow.

Space Rider Digital Model

Felisiak Engineering & Development · 2026

- Reconstructed the ESA Space Rider from public blueprints during a 17-week internship; the internal verification record reports ±6 mm side-view and ±8.4 mm top-view agreement over a 4.88 m body.
- Used scripted geometry audits, protected reference states, rollback branches, and reference-view checks to retain failed approaches and prevent local surface fixes from changing accepted proportions.
- Delivered separate review and animation-ready assets and reduced the browser GLB from 11.6 MB to 639 KB while preserving the declared release structure.
- Limit: reference-led work for Felisiak using public ESA material, not ESA employment, manufacturer CAD, or flight-article validation.

XC48 Abaqus Tensile Twin

Course project · 2026

- Built a Python pipeline from raw force/displacement data to validated material inputs, Abaqus ODB post-processing, and report generation for an XC48 tensile study.
- Matched the retained experiment with R^2=0.9663 on the M2 mesh and verified a 5% pre-fracture kinetic/internal energy bound; the stated limit excludes sole authorship of the full team Abaqus model.

Airfoil & Ground-Effect Methods

Independent studies · 2026

- Implemented a cosine-spaced Hess–Smith panel method for NACA 0012 and compared linear lift against NASA TM-4074; the 160-to-240-panel lift change at 4° was 0.0307%.
- Built a horseshoe/image-vortex ground-effect solver over 14 h/c cases; recovered free-air lift within 0.0034% at h/c=50 and changed 0.258% from 64 to 96 span panels.
- Limit: linear, inviscid, steady methods, without viscous drag, separation, stall, tyres, bodywork, or race-car performance claims.

03

Education

Mechanics, computation, and software

Engineering Degree, Modelling & Computational Mechanics

Sep 2025 — 2027 expected

ESILV · Paris La Defense, France

- Computational fluid dynamics and finite-volume methods, finite-element methods, vehicle dynamics and powertrains, sustainable automotive engineering, structural optimization, machine learning, and digital twins.

Full-time French Language Studies

Apr 2023 — May 2024

French Alliance Beijing · Beijing, China

Bachelor of Computer Science

Sep 2012 — Jul 2016 · GPA 3.33/4.00

Changchun University of Science and Technology · Changchun, China

- Patent: "Mouse with gamepad function" (CN203870574U, issued Oct 8, 2014).
- First Provincial Prize, China Undergraduate Mathematical Contest in Modeling (2014).
- First Prize, 8th CUST ACM Programming Competition (2013).
- National Scholarship, Ministry of Education award, top 0.2% nationwide (2014–2015).
- First Class Scholarship, awarded five times for academic excellence (2012–2015).

04

Core skills

Tools and working languages

Simulation & numerical methods	OpenFOAM, Ansys Fluent, Ansys Mechanical, Star-CCM+, Abaqus, finite-volume fluid simulation, RANS, finite-element analysis, lattice-Boltzmann methods, POD/DMD, thermal-fluid modelling, mesh studies, and convergence studies.
CAD & 3D	CATIA, SolidWorks, and Blender.
Programming	Java/Kotlin, Python, Ruby/Ruby on Rails, C++, C, Rust, TypeScript/React, Node.js, and MATLAB/Simulink.
Languages	French (DELF B2), English (C1), and Chinese (native).

05

Community

Developer education at scale

Core Organizer (part-time)

Nov 2015 — Mar 2023

Google Developer Group Beijing · Beijing, China

- Organized free technical events that regularly drew more than 200 attendees and featured speakers from Google and Silicon Valley.
- Led technical knowledge sharing across mobile development, open-source tools, and emerging engineering topics.